

Materials Download HW - Part 1

Due Date: Monday, 8/31 at 4:59 PM

Please ask instructors/TAs during class or via Piazza if issues occur!

1. Download Windows Powershell (Windows Only)

(WINDOWS ONLY, Skip Step 1 if using Mac / Terminal)

- ☐ **Install** Powershell via Winget: `winget install --id Microsoft.PowerShell --source winget`
- ☐ **Update** Powershell: `winget upgrade --id Microsoft.PowerShell`

2. Setting up a GitHub account

- ☐ Create a GitHub account at github.com (use your USC email)
- ☐ Test if you have Git installed on your machine by typing `git --version` in your **Terminal** (or Windows Powershell). It should be installed, but if there is some error (like git isn't installed), follow [these instructions](#) to install it.
- ☐ Create your SSH key:
 - ☐ Run following command in your terminal (w/ your email to use for GitHub):

```
ssh-keygen -t rsa -b 2048 -C "[your USC email]"
```

You will most likely get a message similar to the following:

```
((base) mi@Michelles-MacBook-Air-2 ~ % ssh-keygen -t rsa -b 2048 -C "jrevilla@usc.edu"  
Generating public/private rsa key pair.  
Enter file in which to save the key (/Users/mi/.ssh/id_rsa):
```

Just hit enter. (The default path that shows up might be different, but don't worry.):

```
[Enter passphrase (empty for no passphrase):  
[Enter same passphrase again:
```

You will then be prompted for a password – while it is optional, you should hit enter to set one up anyways. **Make sure you save it!**

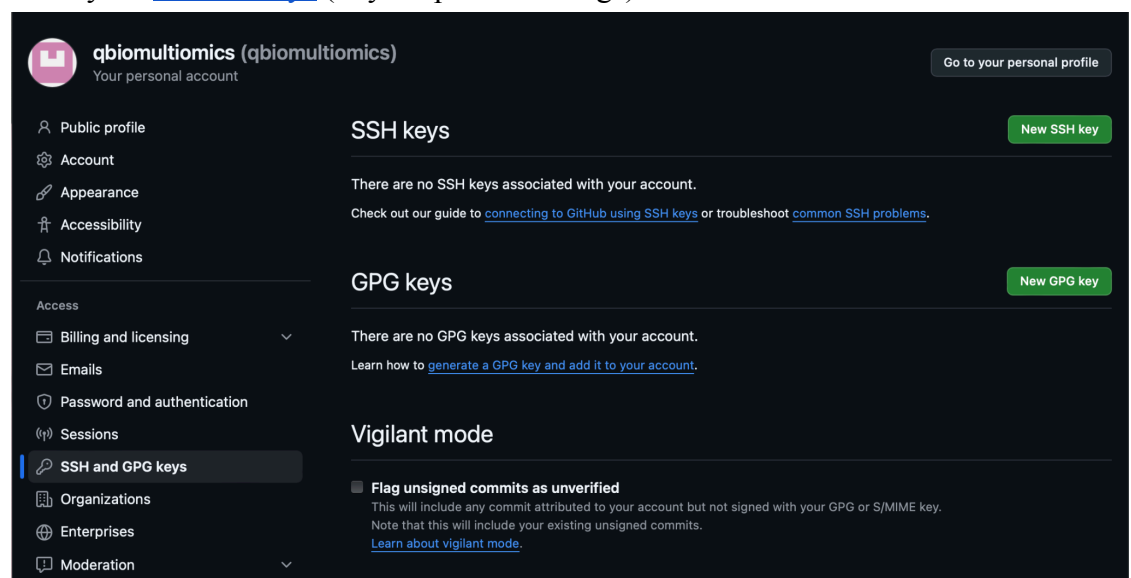
- ☐ Confirm that you get some message that looks like the following:

```
Desktop -- zsh -- 80x17

Your identification has been saved in /Users/mi/.ssh/id_rsa
Your public key has been saved in /Users/mi/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:GxDsWYjX1/4RkvPzhSxi+4B8lp8LDoKK3GOWFKtSxrU jrevilla@usc.edu
The key's randomart image is:
+---[RSA 2048]-----+
|  o.o  .  . |
|  .+.o . = . |
|  o.o . . = o |
|  .. o. o o * . |
|  .o. .So + o +. |
|  +oE. oo* . . |
|  oo o . oo.+ . |
|  oo.* . o .+ |
|  o.+.. . . . |
+---[SHA256]-----+
(base) mi@Michelles-MacBook-Air-2 Desktop %
```

Congratulations! You have created an SSH key. Now, you can move on to adding it to your GitHub account.

- ☐ Add your SSH key to your Git account:
 - ☐ Go to your [GitHub keys](#) (in your profile settings).

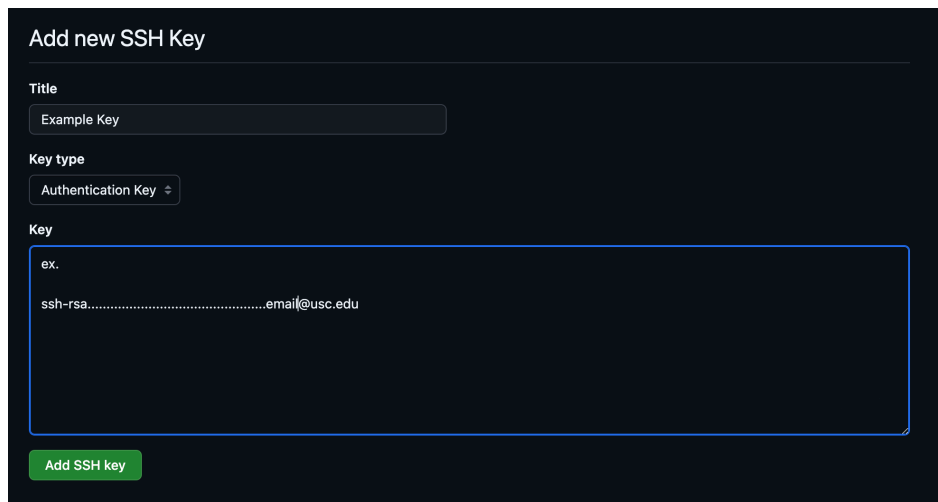


- ☐ Click the button saying “New SSH key” (green, top right)
- ☐ In the title field, name the key (like “[Your name]’s MacBook”)
- ☐ Go back to your terminal and run the following command

```
cat ~/.ssh/id_rsa.pub
```
- ☐ Copy the output, starting with **ssh-rsa** and ending in your email:
 - ☐ Hint: For Mac terminal, `command-c` should work to copy. For Ubuntu, use `ctrl-shift-c` or right click the top banner -> Edit -> Copy

```
[(base) mi@Michelles-MacBook-Air-2 Desktop % cat ~/.ssh/id_rsa.pub  
ssh-rsa  
jrevilla@usc.edu  
(base) mi@Michelles-MacBook-Air-2 Desktop %
```

- ☐ Paste your clipboard into the GitHub key field, then click add key.



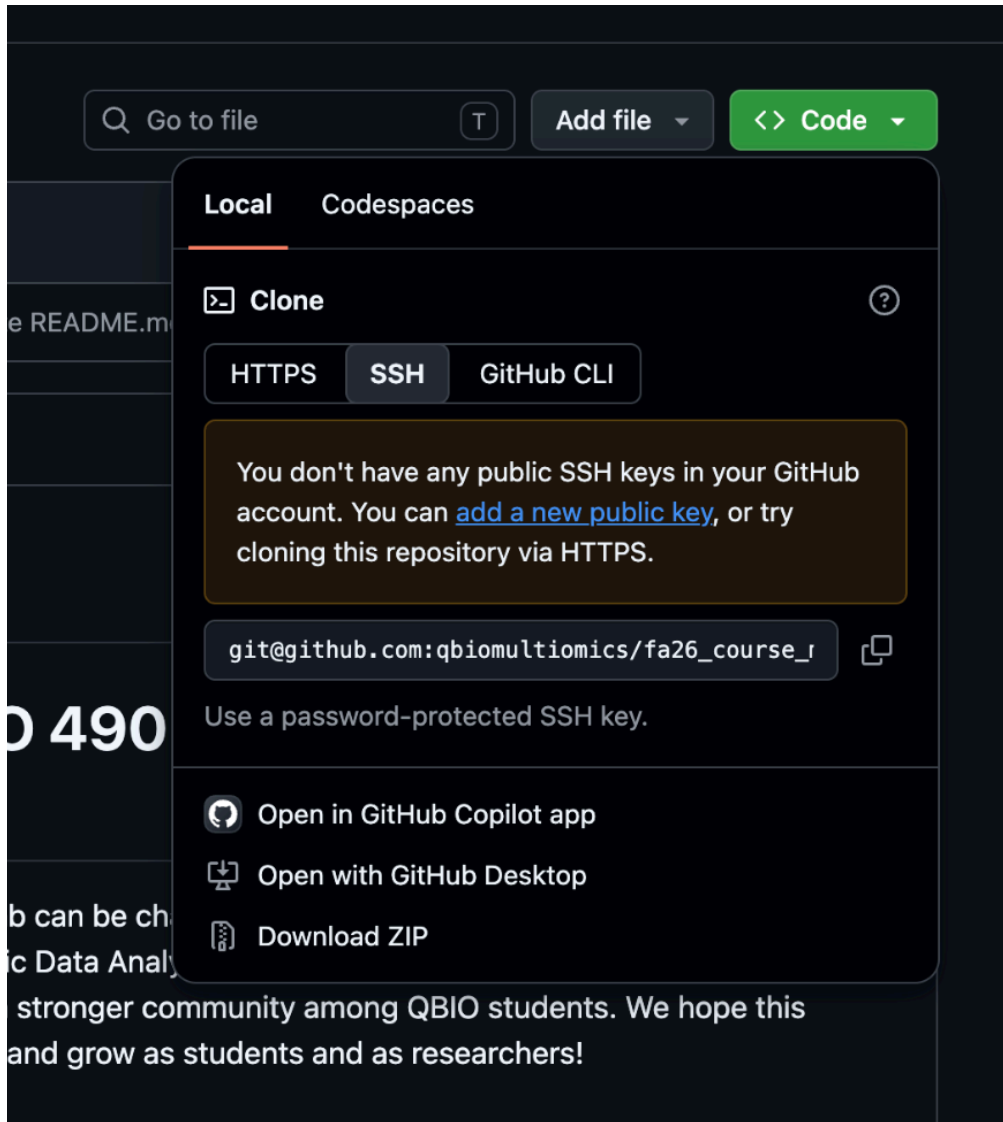
- ☐ Lastly, set up Git on your computer:
 - ☐ Configure your personal information with the following commands in your terminal:

```
git config --global user.name "[your name]"  
git config --global user.email "[your USC email]"
```
 - ☐ (Optional) Set some quality-of-life settings:

```
git config --global color.ui auto  
git config --global core.editor "nano"  
git config --global core.autocrlf input  
git config --global push.default simple
```

3. Clone the course material repository

- ☐ Clone the class repository (fa26_course_materials) to your desktop:
 - ☐ Go to the [fa_26_course_materials | GitHub repository](#).
 - ☐ Click on the green code button (left of the “About” panel), then click copy the SSH link (it should be something like [git@github.com:qbiomultiomics/fa26_course_materials.git](#)).



- ☐ Go to your Terminal and navigate to the location on your computer where you want to keep your work for this course (ex: Desktop, Document, or QBIO folder). If you don't have a QBIO490 folder yet, make one now!

Make sure you know where to find this folder!

- ☐ You can use the bash commands (we will learn more about this soon!) to change folders.
 - ☐ Ex: Mac: `cd /Users/jeanne/Desktop/QBIO` or `cd ~/Desktop/QBIO`
 - ☐ Ex: Windows: `cd /mnt/c/Users/jrevilla/Documents`

Here is an example of using “pwd” (print working directory) to find where you are, “ls” (list) to view what is currently in the folder you are located in, and “cd” (change directory) to move to my QBIO folder on my Desktop. This is where I will keep my materials. If you don't already have a QBIO folder, consider setting one up!

```
[(base) mi@Michelles-MacBook-Air-2 ~ % pwd
/Users/mi
[(base) mi@Michelles-MacBook-Air-2 ~ % ls
Applications          Music
Book1.xlsx             Pictures
Desktop                Public
Documents              Titration Curves.xlsx
Downloads              ch-darwin-arm64
GDCdata                fa23_course_materials
Indicator Spectra.xlsx miniconda3
Library                sp.docx
MANIFEST.txt           zoonotic disease note.txt
Movies
[(base) mi@Michelles-MacBook-Air-2 ~ % cd Desktop
[(base) mi@Michelles-MacBook-Air-2 Desktop % cd QBIO
(base) mi@Michelles-MacBook-Air-2 QBIO %
```

- ☐ Type `git clone <URL you copied>` and hit enter. If you look through your files, you should see the course folder pop up where you pasted.
The image below again shows me using “cd” to move into my QBIO folder, like the above image. If you have already done that, there is no need to repeat it before using git clone.

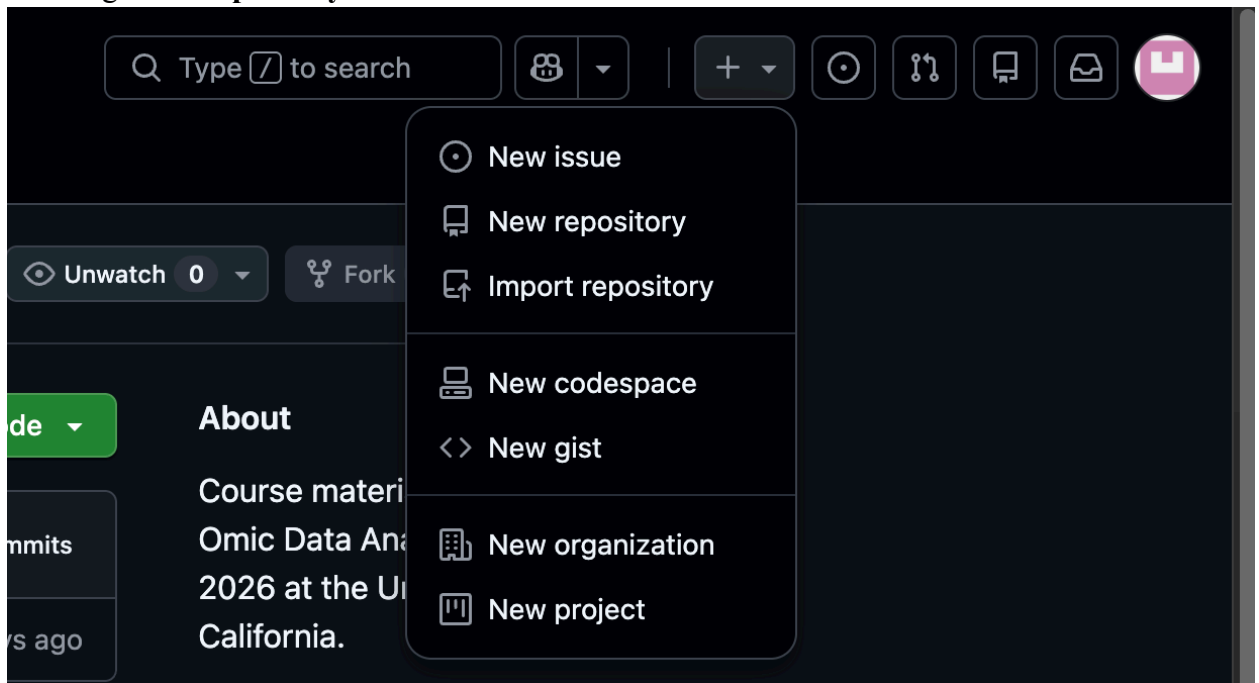
```

[mi@Michelles-MacBook-Air-2 ~ % cd Desktop
[mi@Michelles-MacBook-Air-2 Desktop % cd QBIO
[mi@Michelles-MacBook-Air-2 QBIO % git clone git@github.com:]
qbiomultimomics/sp26_course_materials.git
Cloning into 'sp26_course_materials'...

```

4. Create your own repository and add to GitHub

- ☐ Create a new repository in GitHub by hitting the plus arrow in the top right corner and choosing **New Repository**.



- ☐ Name your personal repository for this class (**qbio_490_name**), add a description and/or readme (optional), and *make sure to mark your repository as public* so we can see your homework! Choose any template for your .gitignore (we're going to rewrite it in a second).

Create a new repository [\(Preview\)](#) [Switch back to classic experience](#)

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1

General

Owner *

jmfrevilla

 /

qbio_490_jeannerevilla

qbio_490_jeannerevilla is available.

Great repository names are short and memorable. How about **didactic-fishstick**?

Description

0 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

On

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

AL

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

- ☐ After creating the repository, click on the green code button (left of the “About” panel), then click copy the SSH link

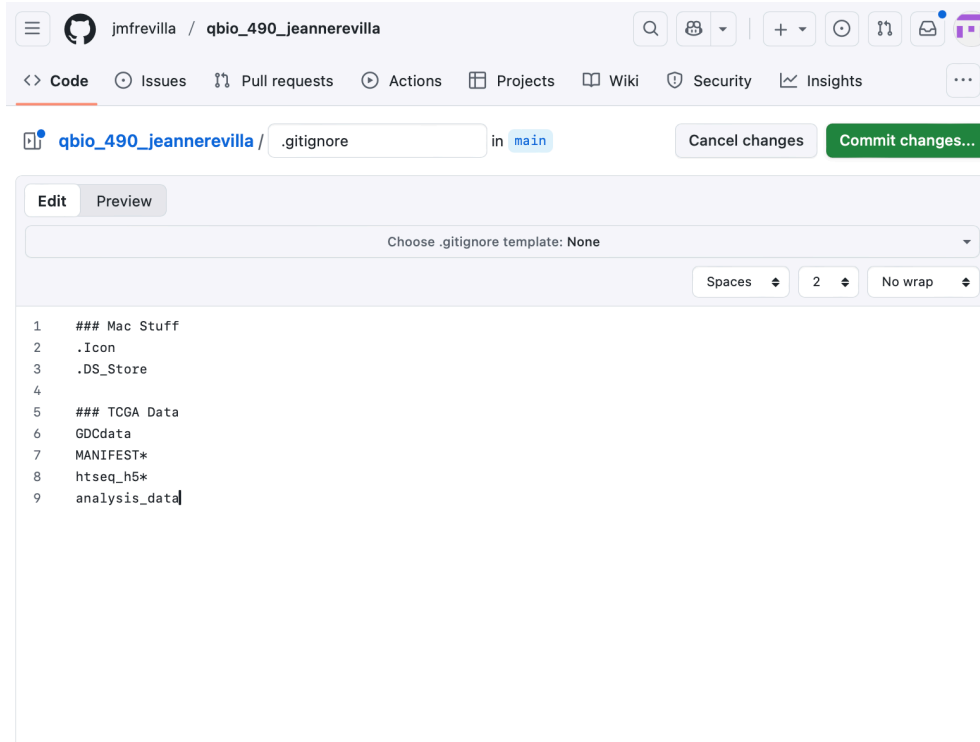
The screenshot shows the GitHub repository page for 'qbio_490_jeannerevilla' under the user 'jmfrevilla'. The repository is public and has no description, website, or topics provided. The 'Code' button is highlighted, and the dropdown menu is open, showing options to clone the repository using HTTPS, SSH, or GitHub CLI. The SSH link is displayed as 'git@github.com:jmfrevilla/qbio_490_jeanne:'. The 'About' panel on the right shows the repository has 0 stars, 0 watching, and 0 forks. The 'Releases' and 'Packages' sections also show no published items.

- ☐ Go to your terminal and navigate to the location on your computer where you want to keep your work for this course (ex: Desktop, Document, or QBIO folder).

- ☐ Type `git clone <URL you copied>` and hit enter. You should see the folder pop up where you pasted.
- ☐ In GitHub, click on `.gitignore`, then click on edit. Delete the current contents and type:

```
### Mac Stuff
.Icon
.DS_Store

### TCGA Data
GDCdata
MANIFEST*
htseq_h5*
analysis_data
```



Click the green “Commit changes” button. If prompted with another commit changes window, click on the green “Commit changes” button again.

Once all these steps are done, you should be good to go on your local machine!

5. Setup your CARC OnDemand RStudio environment

If you do not have CARC access yet, please contact Joshua and/or Chris.

1. Ensure you are either on the **USC Secure Wireless Wifi** or using the USC VPN service if you are not on the USC wifi. Instructions can be found [here](#).
2. Go to ondemand.carc.usc.edu
3. Click RStudio Server and use the following arguments (memory and hours may be changed as needed):

RStudio Server

This app will launch [RStudio Server](#), an IDE for [R](#), on a compute node.

Cluster

discovery

Cluster to submit the job to.

RStudio Server version

2025.05.1+513

Software environment

modules

R version

4.5.0

Additional software modules to load (optional)

cmake

Space-separated list of additional Lmod software modules to load (from same compiler stack as R version).

Project account (optional)

Project account to charge resources to. If blank, default account is used.

Partition

main

Partition to submit the job to. For the Discovery cluster, see [Discovery Resource Overview](#) for more information on resources allocated to each partition.

Number of CPUs

2

Number of CPU cores to allocate.

Memory (GB)

4

Amount of memory to allocate. If blank, 2 GB of memory per CPU core is allocated.

Constraint (optional)

Constraint options (node features) (e.g., epyc-7513).

Number of hours

2

Amount of time to allocate. (On Discovery cluster: 1 hr max for **debug** partition.)

Reservation (optional)

Name of reservation. If multiple, use a comma-separated list.

☐ I would like to receive an email when the session starts

Launch

* The RStudio Server session data for this session can be accessed under the [data root directory](#).

4. Click launch and wait for the status of the job to change to "Running".

5. Copy the password and click “Connect to RStudio Server” then input the given username and password.

RStudio Server (2057095) **1 node** | **2 cores** | **Running**

Host: `>_ a03-03.hpc.usc.edu` ✕ Cancel

Created at: 2025-09-02 15:23:11 PDT

Time Remaining: 1 hour and 59 minutes

Session ID: 82e9e0a2-cadf-4e66-b76e-60719d3f7dbf

Problems with this session? [Submit support ticket](#)

Use the following credentials to log in to RStudio Server:

Username: jrevilla
Password: AcepvNNnVOTUavdb

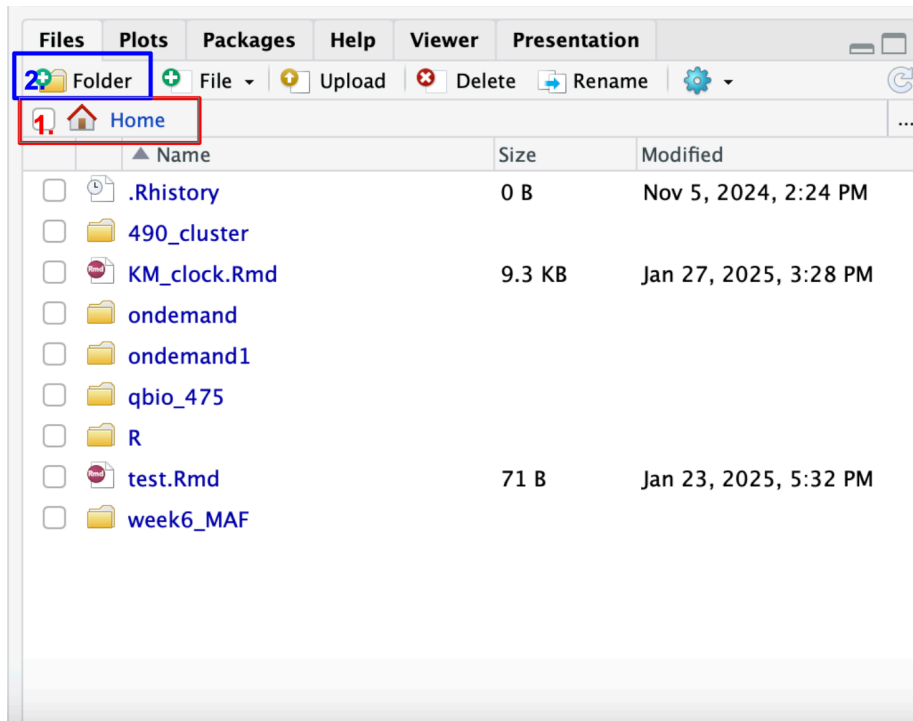
® **Connect to RStudio Server**

6. You should see a IDE that looks something like this:

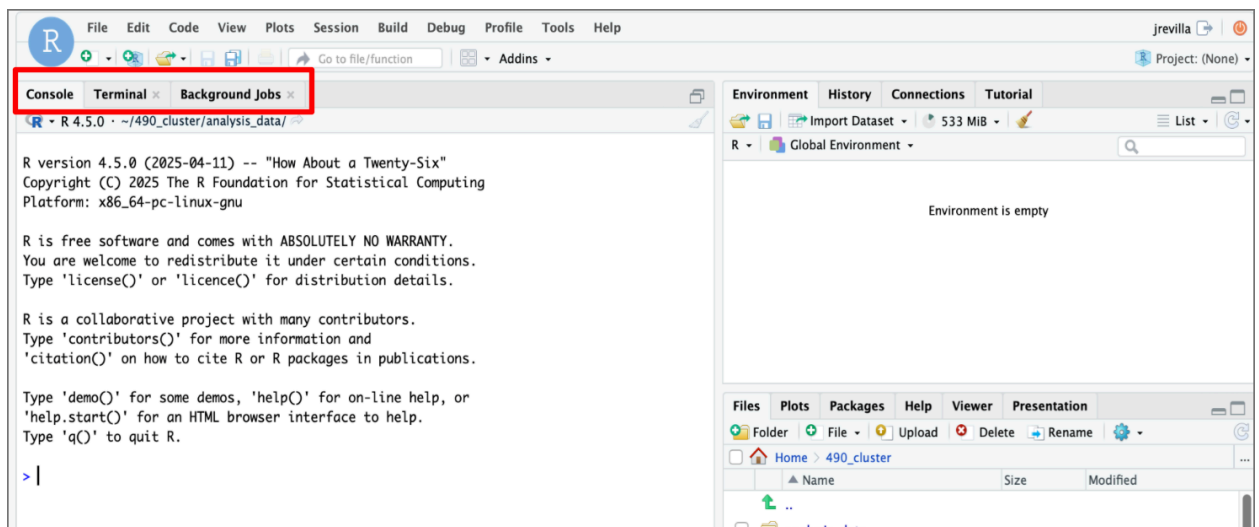
The screenshot displays the RStudio IDE interface. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. The left sidebar contains the Console, Terminal, and Background Jobs panels. The main workspace is divided into two panes: the top pane shows the Environment, History, Connections, and Tutorial tabs, with the Environment tab active displaying 'Environment is empty'; the bottom pane shows the Files, Plots, Packages, Help, Viewer, and Presentation tabs, with the Files tab active displaying a file explorer view of the home directory. The file explorer lists various files and folders, including .Rhistory, analysis_data, bar92.DP8MP4BIMAC2HET75.hapFem.minimal.corrected.vcf.gz, distance_analysis, hell_focal_samples.vcf.gz, metadata, ondemand, qbio_475, R, Spatial Analysis, Transcriptomics_Tutorial.Rmd, and Zombie-Barcodes.

Name	Size	Modified
.Rhistory	400 B	Sep 10, 2024, 4:45 PM
analysis_data		
bar92.DP8MP4BIMAC2HET75.hapFem.minimal.corrected.vcf.gz	437.6 MB	Nov 2, 2023, 12:54 PM
bar92.DP8MP4BIMAC2HET75.hapFem.minimal.corrected.vcf.gz.tbi	187.1 KB	Nov 2, 2023, 1:42 PM
distance_analysis		
hell_focal_samples.vcf.gz	24 MB	Nov 2, 2023, 1:43 PM
hell_focal_samples.vcf.gz.tbi	11.4 KB	Nov 2, 2023, 1:43 PM
metadata		
ondemand		
qbio_475		
R		
Spatial Analysis		
Transcriptomics_Tutorial.Rmd	607 B	Sep 10, 2024, 4:45 PM
Zombie-Barcodes		

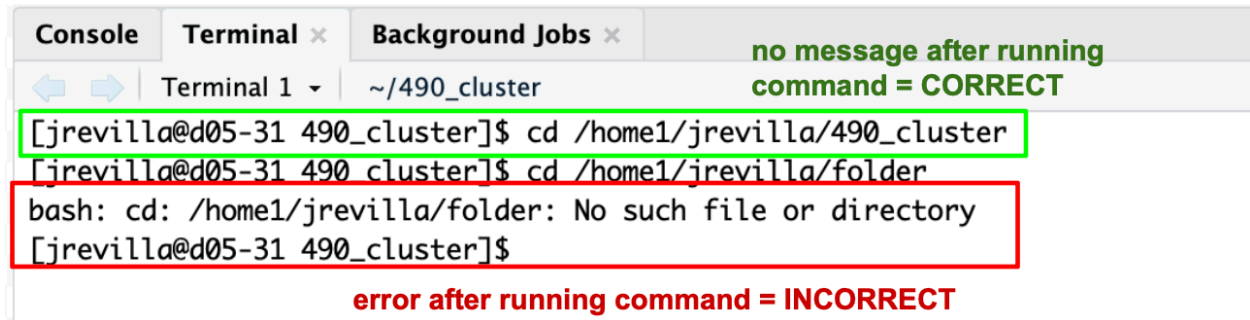
7. In the bottom right graphical pane, click “Home”, then “New Folder”. Create a new folder called **490_cluster**.



8. On the lefthand panel, navigate to the Terminal tab.



9. Type `cd /home1/[your USC username]/490_cluster`, using **your USC email username** (without the @usc.edu). If the command runs without error, you have done this step successfully.



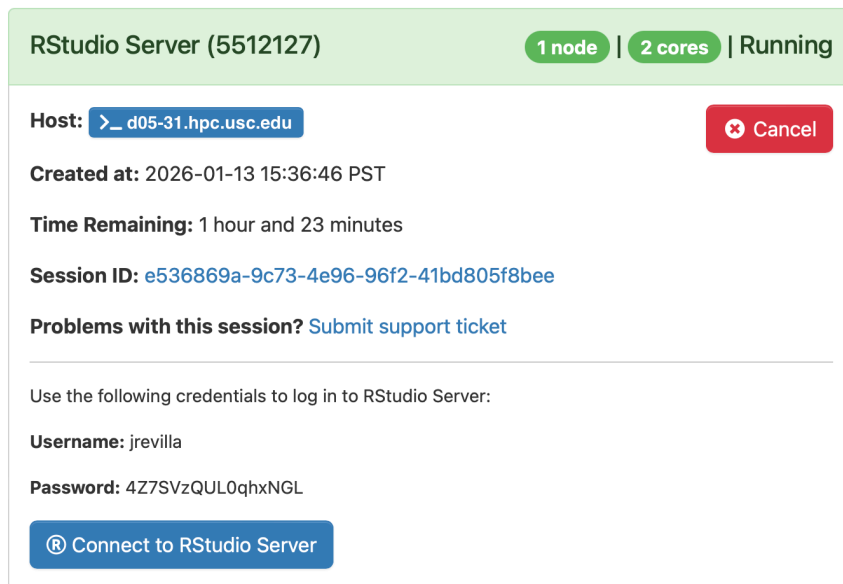
The screenshot shows a terminal window with tabs for Console, Terminal, and Background Jobs. The Terminal tab is active, showing the command prompt `[jrevilla@d05-31 490_cluster]$`. The first command `cd /home1/jrevilla/490_cluster` is highlighted with a green box, and the second command `cd /home1/jrevilla/folder` is highlighted with a red box. The output of the second command is `bash: cd: /home1/jrevilla/folder: No such file or directory`. Below the terminal window, the text `error after running command = INCORRECT` is displayed in red.

```
[jrevilla@d05-31 490_cluster]$ cd /home1/jrevilla/490_cluster
[jrevilla@d05-31 490_cluster]$ cd /home1/jrevilla/folder
bash: cd: /home1/jrevilla/folder: No such file or directory
[jrevilla@d05-31 490_cluster]$
```

error after running command = INCORRECT

If you get an error, that's okay and you can continue the assignment. However, it may make navigating your files more difficult later on.

10. Return to “My Interactive Sessions” on CARC OnDemand and shut down your current session by hitting cancel. ***You should always do this whenever you are done with your session!!!***



The screenshot shows the RStudio Server interface for a session named "RStudio Server (5512127)". The session is running on 1 node with 2 cores. The host is `>_ d05-31.hpc.usc.edu`. The session was created at 2026-01-13 15:36:46 PST and has 1 hour and 23 minutes remaining. The session ID is `e536869a-9c73-4e96-96f2-41bd805f8bee`. There is a link to submit a support ticket. The interface also displays the username `jrevilla` and the password `4Z7SVzQUL0qhXNGL`. A "Cancel" button is visible in the top right corner.

RStudio Server (5512127) 1 node | 2 cores | Running

Host: `>_ d05-31.hpc.usc.edu` Cancel

Created at: 2026-01-13 15:36:46 PST

Time Remaining: 1 hour and 23 minutes

Session ID: `e536869a-9c73-4e96-96f2-41bd805f8bee`

Problems with this session? [Submit support ticket](#)

Use the following credentials to log in to RStudio Server:

Username: `jrevilla`

Password: `4Z7SVzQUL0qhXNGL`

Connect to RStudio Server

HW to turn in for credit:

Copy the URL link to your personal `qbio_490_name` folder and submit it to the Brightspace assignment.